MIL-M-28787/221A(EC)
8 June 1982
SUPERSEDING
MIL-M-28787/221(NAVY)
29 July 1974

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

AMPLIFIER, PHASE SHIFT, ANALOG, KEY CODE LEE

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic phase shift amplifier module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.983 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be LEE.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 20 contact pins positioned A21 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/221-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

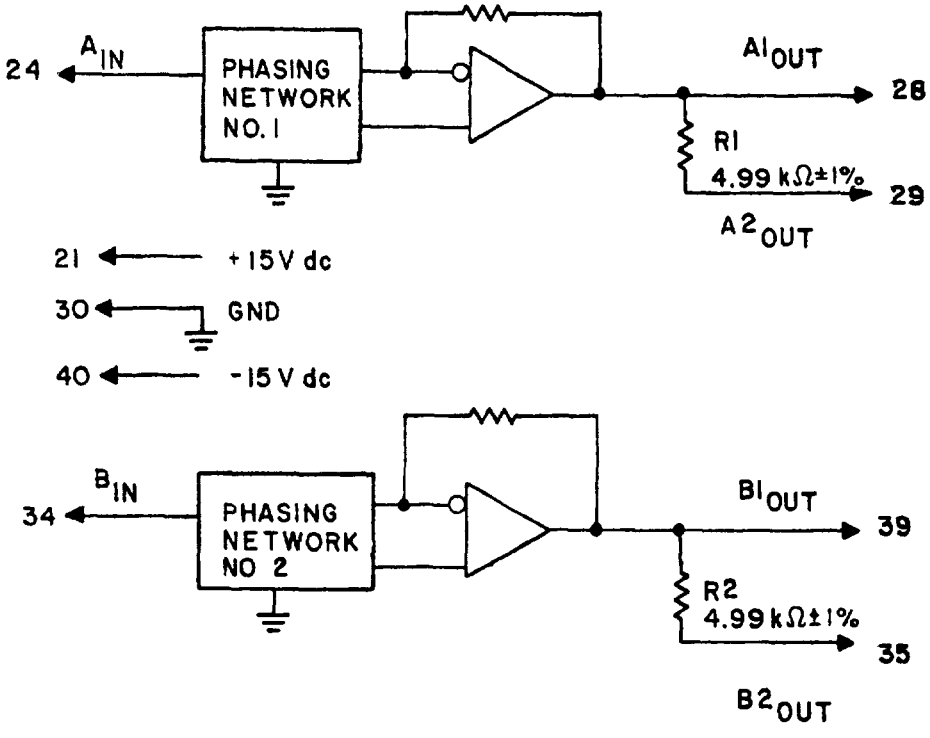
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES
1. Pins 22, 23, 25-27, 31-33, 36-38 are not used.
2. Pins 1-20, if supplied, are not used
3. Each unused pin shall be isolated from all other pins.

FIGURE 1. LEE module function diagram, phase shift amplifier.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1, A2	Isolation	II	S	10	∞	1	∞	10	∞	1	∞	MΩ
B1, B2	Output short circuit	III	Q	1	---	1	---	1	---	1	---	Minute
C1, C2	Stress	IV	S	6.72	7.42	6.72	7.42	6.72	7.42	6.72	7.42	V
D1, D2	Resistance check	V	X	2.4	2.6	2.4	2.6	2.4	2.6	2.4	2.6	V
E1-E6	Gain accuracy (3 to 100 kHz)	VI	X	0.95	1.05	0.95	1.05	0.95	1.05	0.95	1.05	V
E7, E8	Gain accuracy (100 kHz to 1 MHz)	VI	S	0.00	1.25	0.00	1.25	0.00	1.25	0.00	1.25	V
F1-F5	Phase accuracy	VII	X	87	93	87	93	85	95	85	95	Degree
G1, G2	Output offset voltage	VIII	S	-25	25	-25	25	-25	30	-25	30	mV
H1, H2	Output noise	IX	S	0	20	0	20	0	30	0	30	µV
J1	Input impedance check	X	S	1.08	1.20	1.08	1.20	1.08	1.20	1.08	1.20	V
J2	Input impedance check	X	S	0.95	1.05	0.95	1.05	0.95	1.05	0.95	1 05	V
K1, K2	Output impedance check	XI	S	14.25	31 5	14.25	31 5	14.25	31.5	14.25	31.5	mV
L1, L2	Quiescent power supply current test	XII	Q	3.5	16.0	3.5	16.0	3	18	3	18	mA
M1, M2	Input common mode voltage swing	XIII	S	6.65	7.35	6.65	7 35	6.65	7.35	6 65	7.35	V
N1-N4	Power supply ripple rejection	XIV	Q	0	4	0	4	0	5	0	5	mV
P1, P2	Power supply transient rise time	XV	S	25	75	25	75	25	75	25	75	µs

1/ "X" parameters shall be 100 percent tested during module acceptance tests

2/ "S" parameters shall be sample acceptance tested (one percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through P shall be performed in any order with the following exceptions. If test conditions A, B or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through P in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through P, shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through P shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through P shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Military Specification MIL-M-28787/221 supersedes Department of the Navy, Naval Electronic Systems Command Drawing No. 28787-0102-876. Part number M28787/221-1 supersedes part numbers 0102-876 and M28787/221.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits of table I.

Supply voltage	±15.0 Vdc ±10 percent
Bandwidth (1.0 Vrms input)	Output voltage 1.00 ±0.05 Vrms from 3 to 100 kHz, 0 to 1.25 Vrms from 100 kHz to 1 MHz
Maximum output signal	10 V peak for load $\geq$2 kΩ

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Relative phase shift	90° ±5°C, 3 to 18 kHz
Quiescent power (no load, no signal)	0.33 W maximum
Impedance (at 18 kHz)	Circuit A, 3.2 kΩ ±5 percent Circuit B, 2.37 kΩ ±5 percent
Output impedance	10Ω maximum

6.5 Module function. The module contains two phase shift networks. The phase shift at the outputs relative to each other, with inputs commoned, is 90° ±3°C over the frequency range 3 to 18 kHz at 25°C. The networks each have unity gain over this bandwidth.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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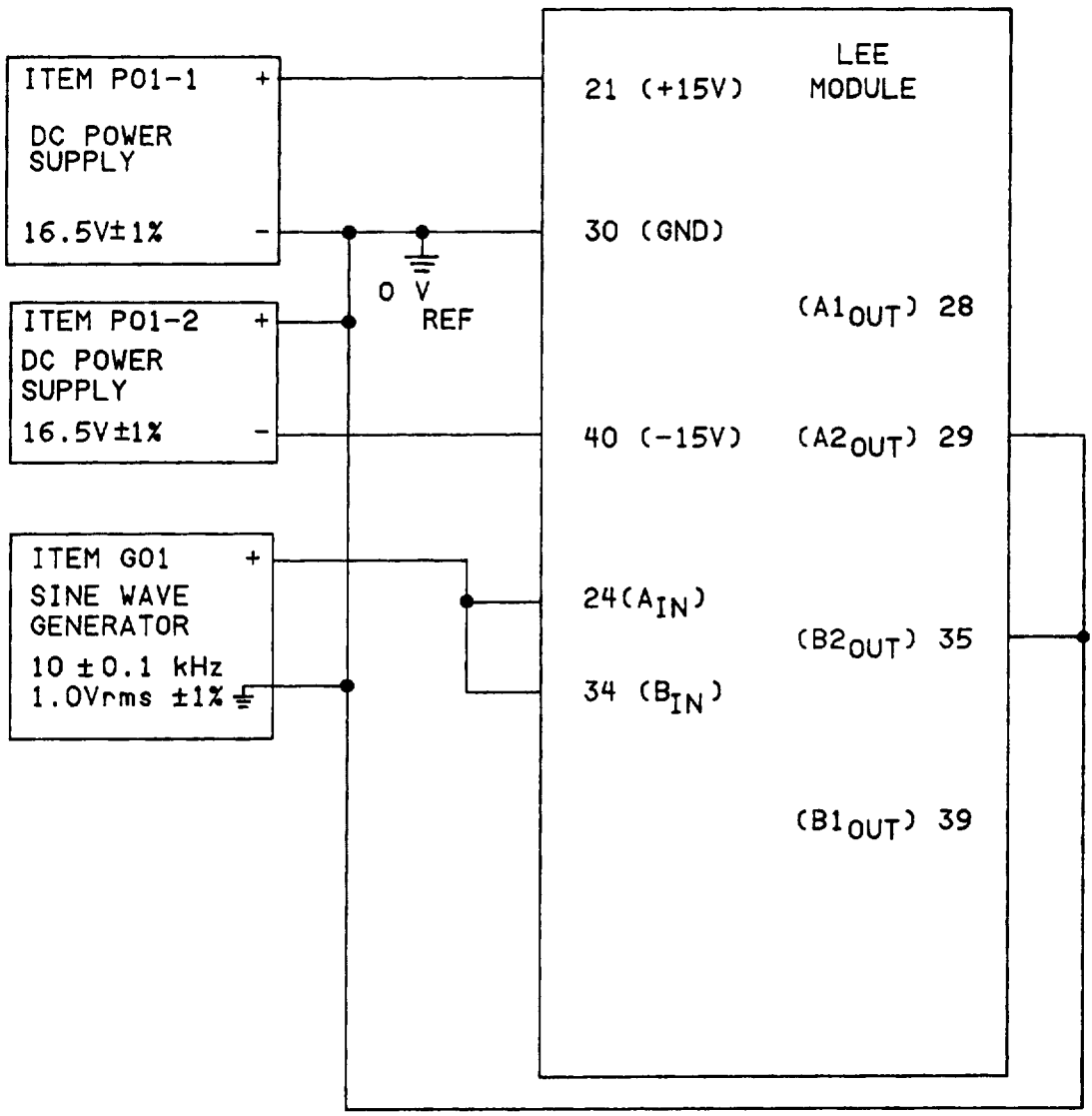


FIGURE 2 Operating temperature, component temperature and life test circuit.

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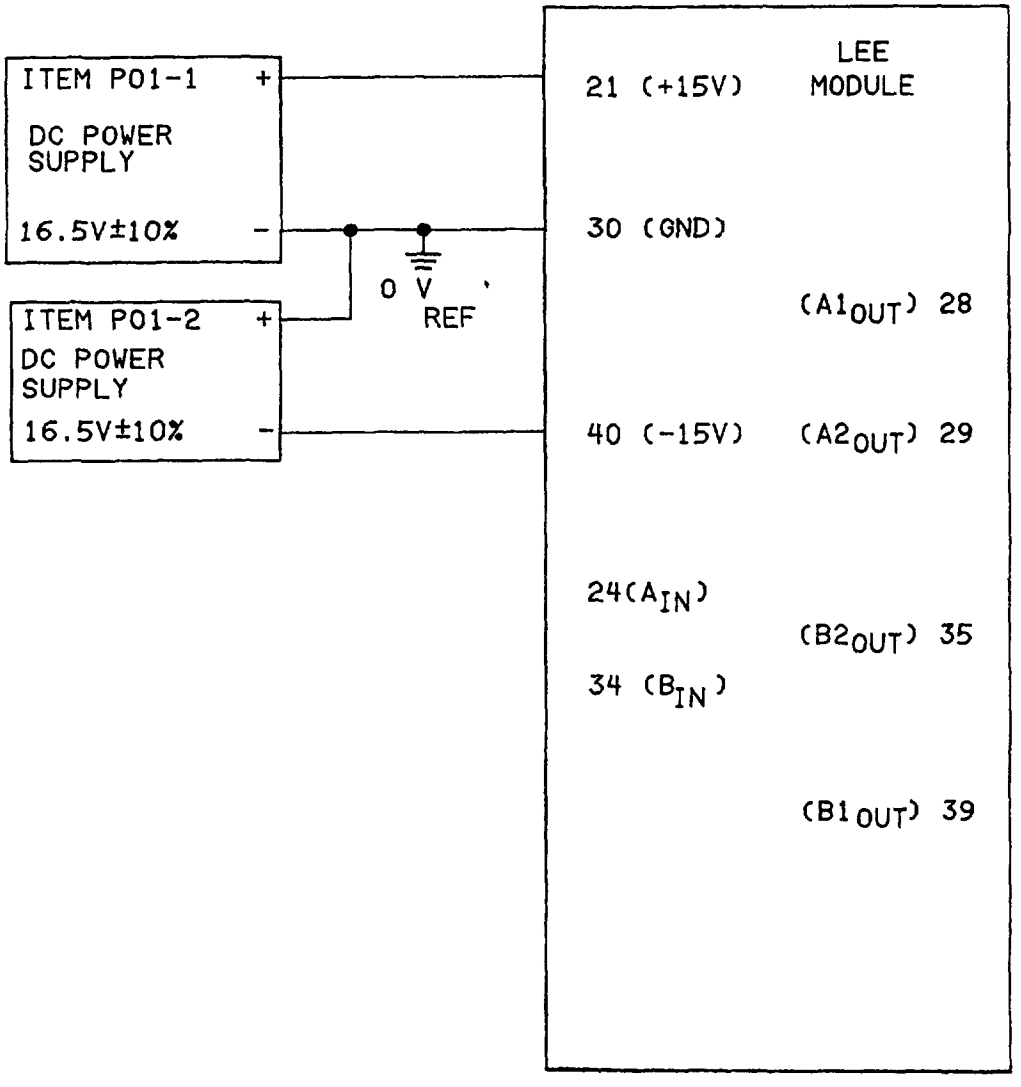


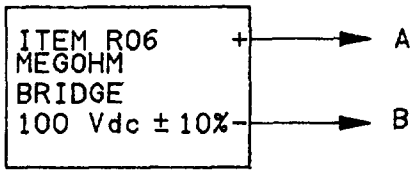
FIGURE 3. Vibration test circuit.

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TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	All pins	Frame	10	∞	MΩ	4
A2	Isolation	21,24,28,29,30, 34,35,39,40	22,23,25,26,27, 31,32,33,36,37, 38	10	∞	MΩ	4

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE On test condition A (when measuring to frame) penetrate the anodized finish of the module frame with a sharply pointed probe connected to the negative terminal.

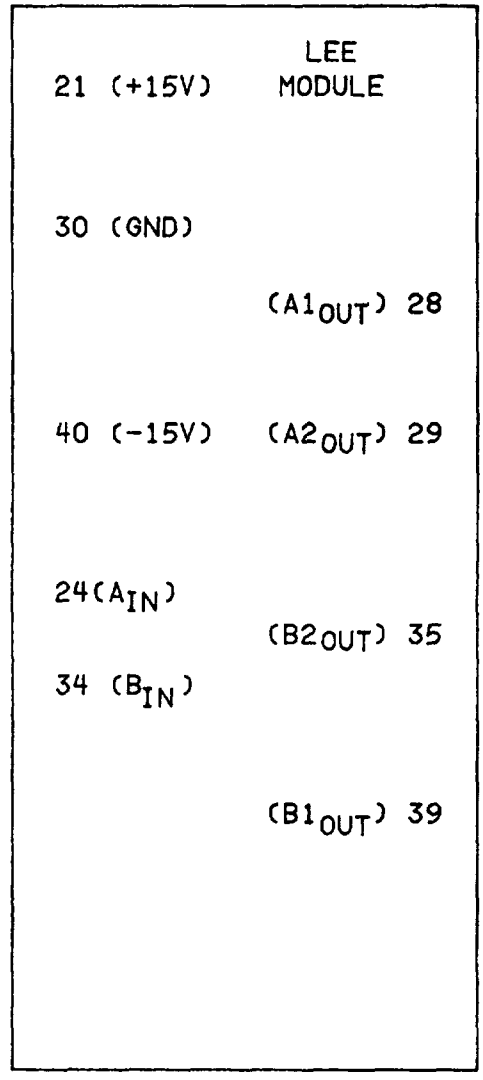


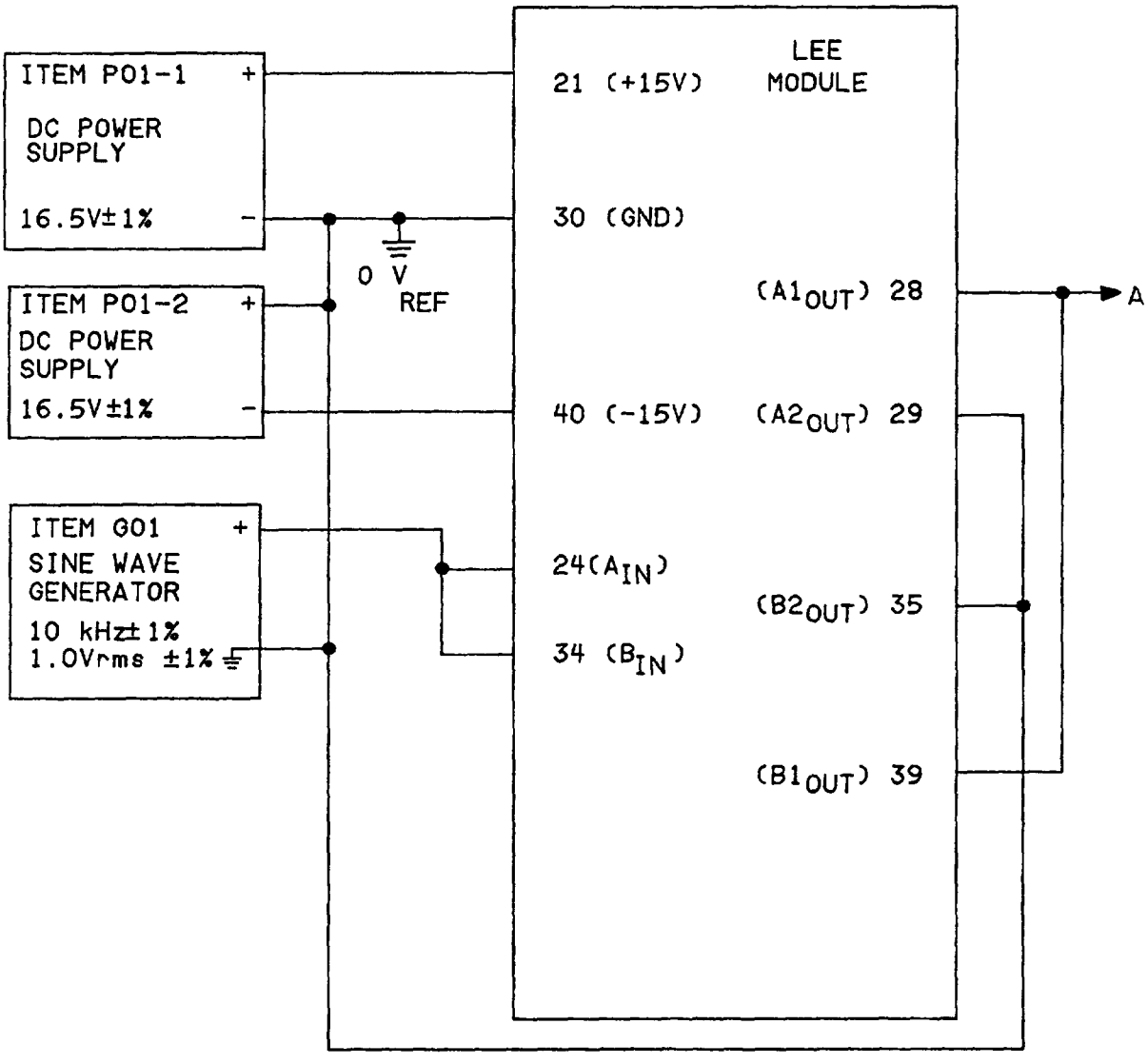
FIGURE 4. Isolation test circuit for table II.

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TABLE III. Output short circuit test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
B1	Output short circuit	1	---	Minute	5
B2	Output short circuit	1	---	Minute	5

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CAUTION

If these tests are successively repeated, a time interval of 10 seconds minimum between tests shall be allowed to preclude detrimental effects due to module heating.

NOTE Connect point A to 0 V reference for one minute minimum.

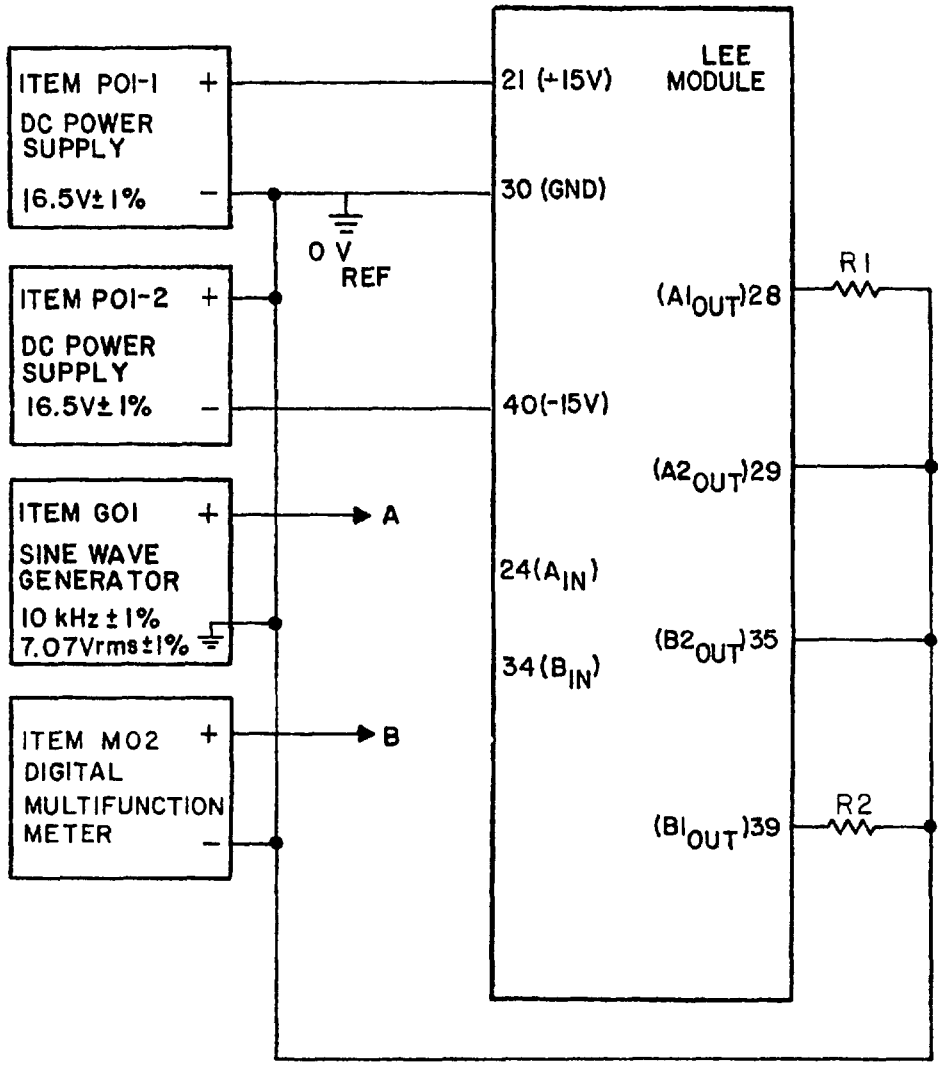
FIGURE 5. Output short circuit test circuit for table III.

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TABLE IV. Stress test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Stress	24	28	6.72	7.42	V	6
C2	Stress	34	39	6.72	7.42	V	6

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CAUTION

If these tests are successively repeated a time interval of 10 seconds minimum between tests shall be allowed to preclude detrimental effects due to module heating.

NOTE: R1,R2 = 2.37 kΩ ±1%, 1/8 W.

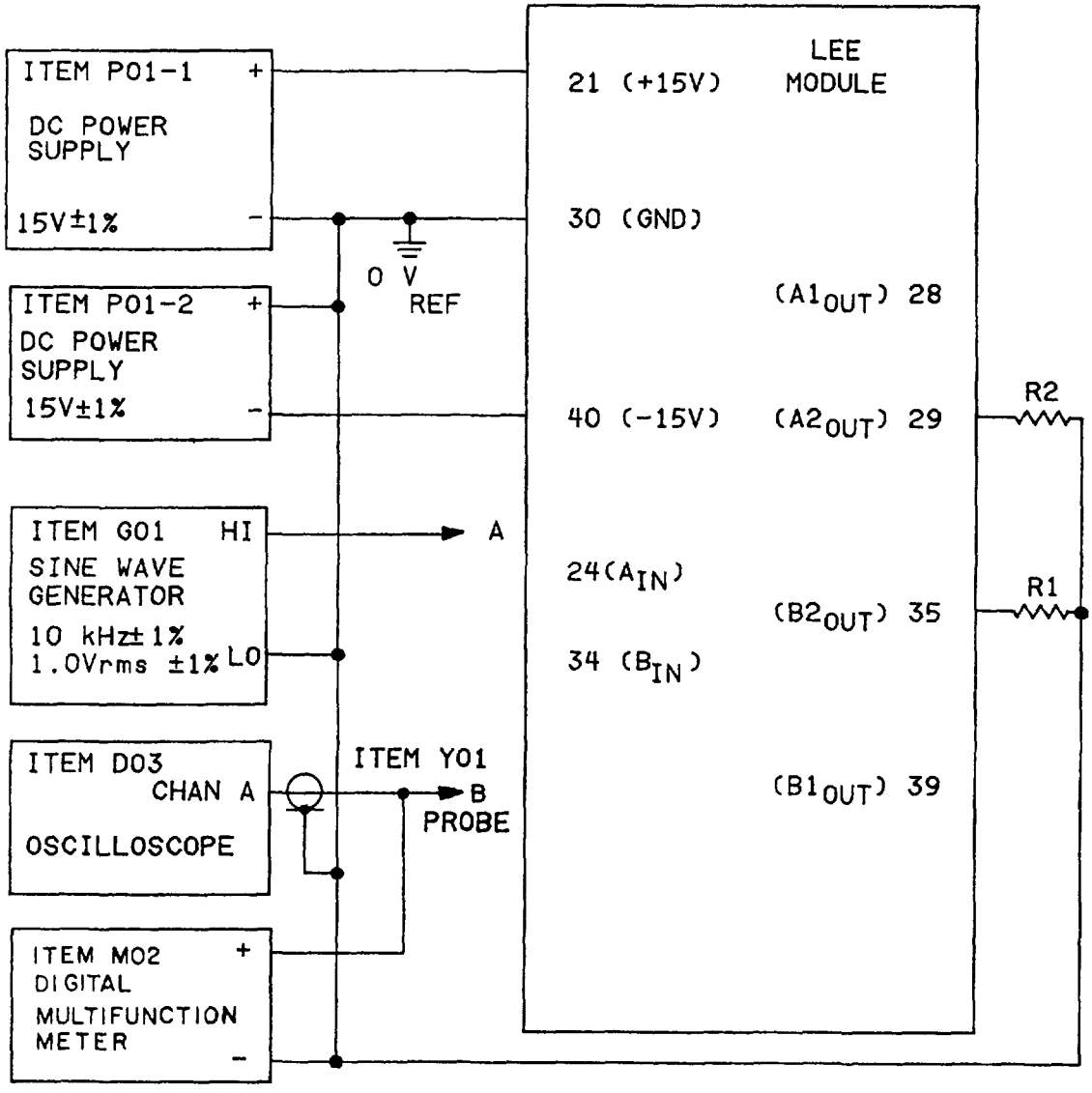
FIGURE 6. Stress test circuit for table IV.

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TABLE V. Resistance check.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	Resistance check	24	29	2.4	2.6	V	7
D2	Resistance check	34	35	2.4	2.6	V	7

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NOTES.
1 R1, R2 = 4.99 kΩ ±1%, 1/8 W.
2 Before test condition D1, connect point A to pin 24, point B to pin 28 - adjust G01 to give 5 V rms ±1% on item W03
3 Before test condition D2, connect point A to pin 34, point B to pin 39 - adjust G01 to give 5 V rms ±1% on item M02.

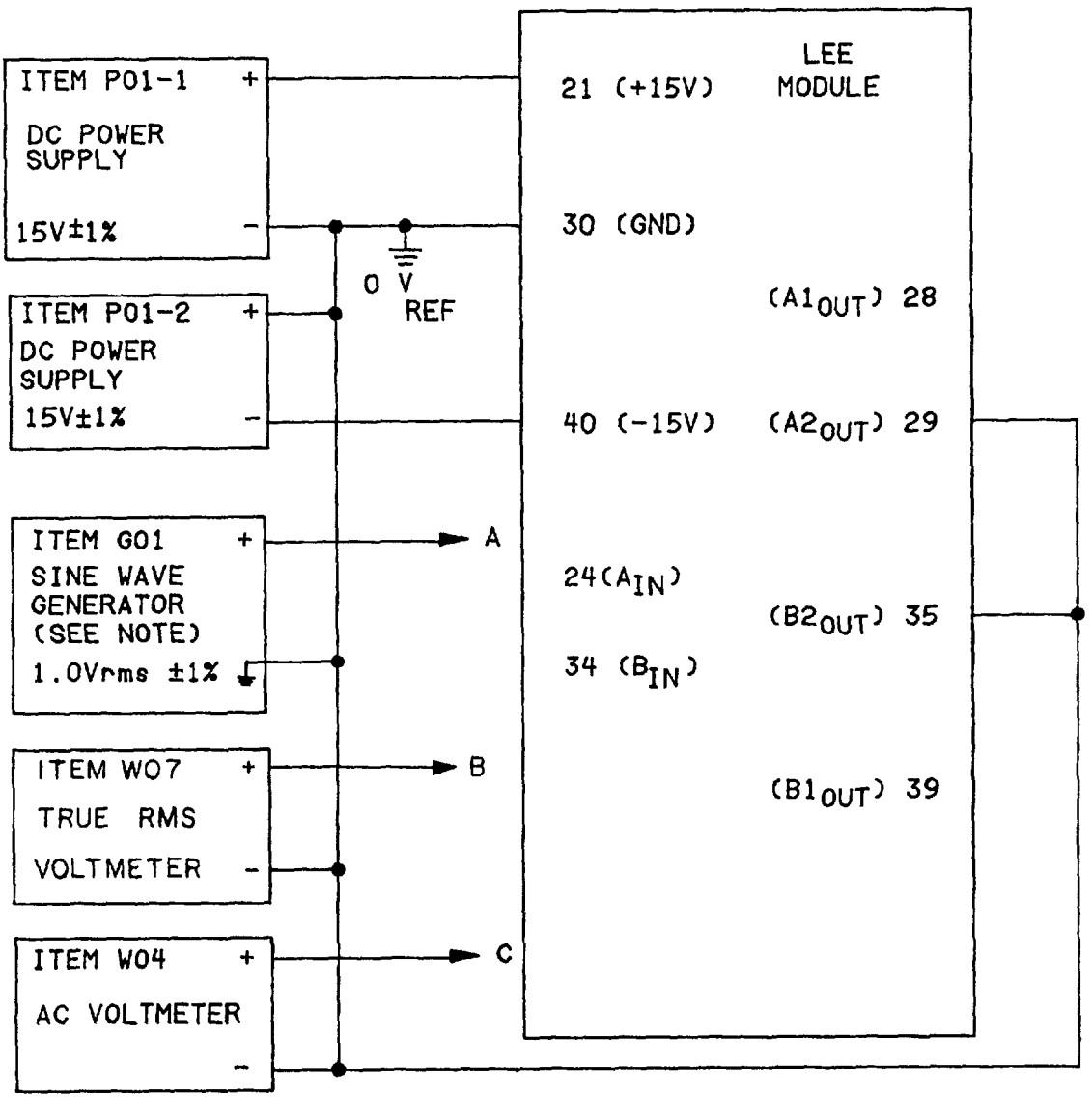
FIGURE 7. Resistance check test circuit for table V.

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TABLE VI. Gain accuracy check test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E1	Gain accuracy (3 to 100 kHz)	24	28	Open	0.95	1.05	V	8
E2	↓	24	28	↓	↓	↓	↓	↓
E3		24	28					
E4		34	39					
E5		34	39					
E6		34	39					
E7	Gain accuracy (100 kHz to 1 MHz)	24	Open	28	0.00	1.25		
E8	Gain accuracy (100 kHz to 1 MHz)	34	Open	39	0.00	1.25		

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NOTE: Item G01 setup:
3 kHz ±1% for test conditions E1, E4.
18 kHz ±1% for test conditions E2, E5.
Tune from 3 to 100 kHz, observe that item W07 reads 1 V ±5% over the band for test conditions E3, E6.
Tune from 100 kHz to 1 MHz, observe that item W04 reads from 0 to 1.25 V over the band for test conditions E7, E8.

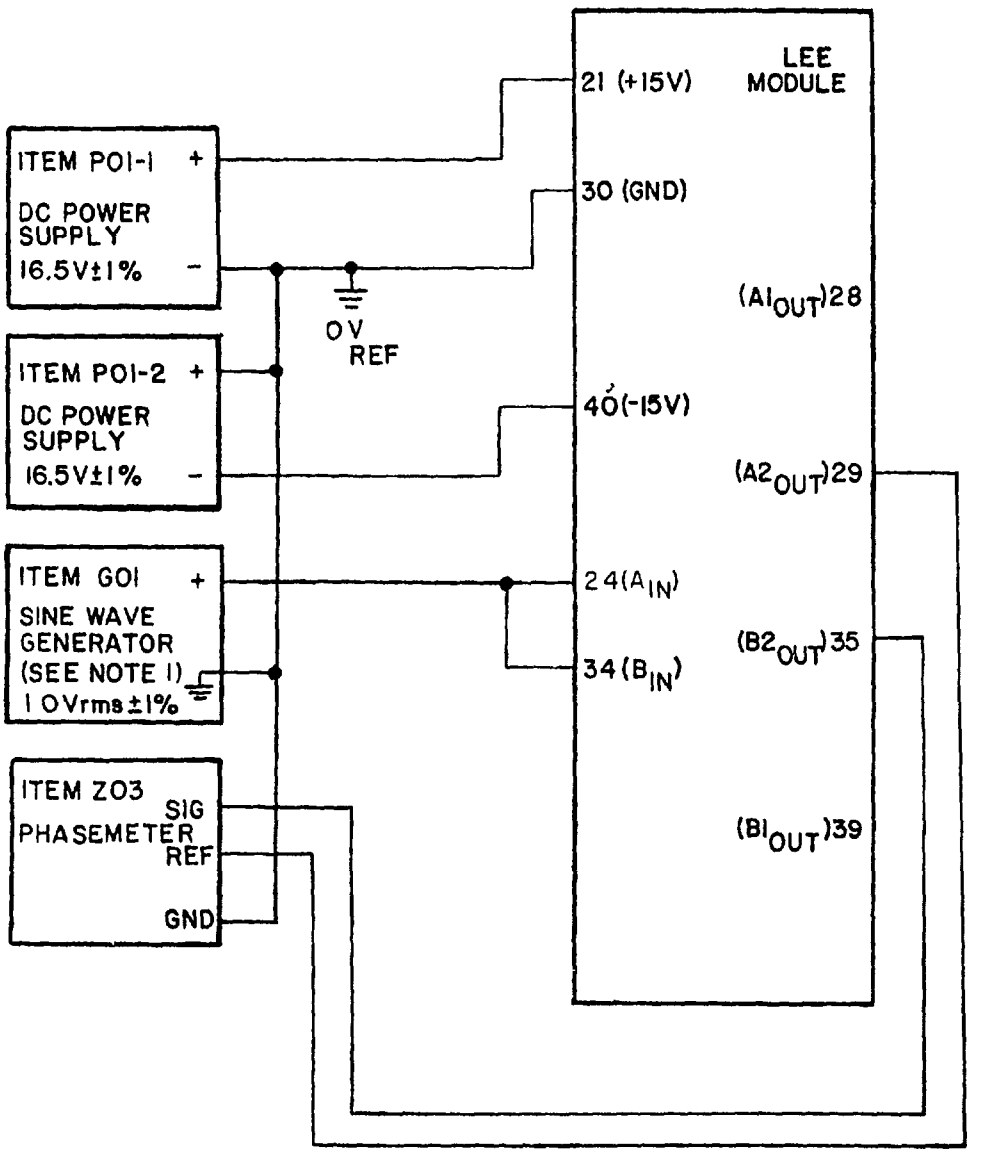
FIGURE 8. Gain accuracy check test circuit for table VI.

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TABLE VII. Phase accuracy test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
F1	Phase accuracy (3 kHz ±1%)	87	93	Degree	9
F2	Phase accuracy (18 kHz ±1%)	↓	↓	↓	↓
F3	Phase accuracy (4.5 kHz ±1%)				
F4	Phase accuracy (7.5 kHz ±1%)				
F5	Phase accuracy (15 kHz ±1%)				

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NOTES:
1. Set item GOI as follows:
   3 kHz ±1% for test condition F1
   18 kHz ±1% for test condition F2.
   4 5 kHz and tune for maximum phase shift as indicated on item ZO3 for test condition F3.
   7.5 kHz and tune for minimum phase shift as indicated on item ZO3 for test condition F4.
   15 kHz and tune for maximum phase shift as indicated on item ZO3 for test condition F5.
2. The leads going from the module under test to the reference terminal and from the module under test to the signal terminal of item ZO3 shall be equal within 0.5 inch (12.7 mm).

FIGURE 9. Phase accuracy test circuit for table VII.

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TABLE VIII. Output offset voltage test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
G1	Output offset voltage	28	-25.0	25.0	mV	10
G2	Output offset voltage	39	-25.0	25.0	mV	10

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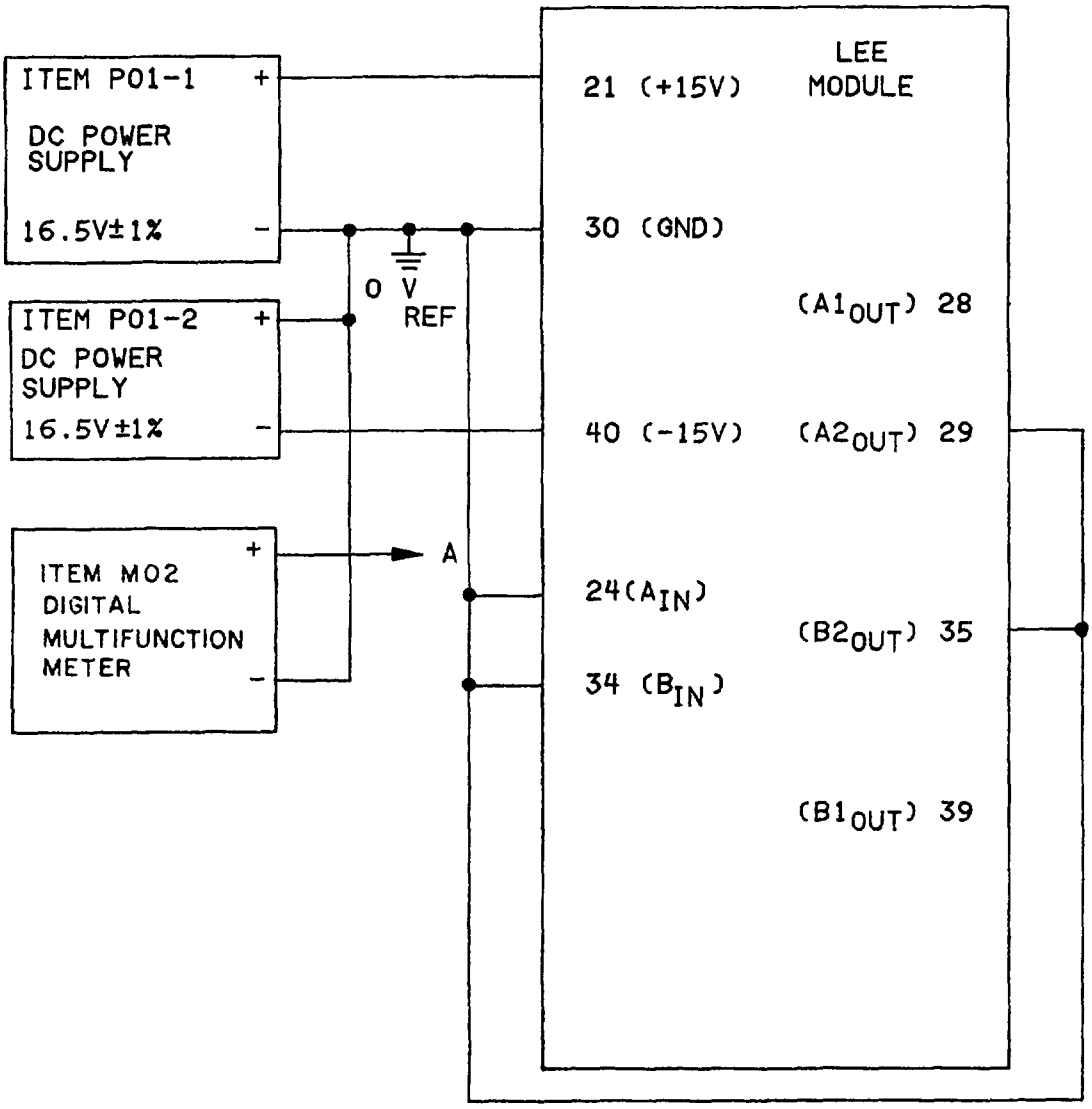


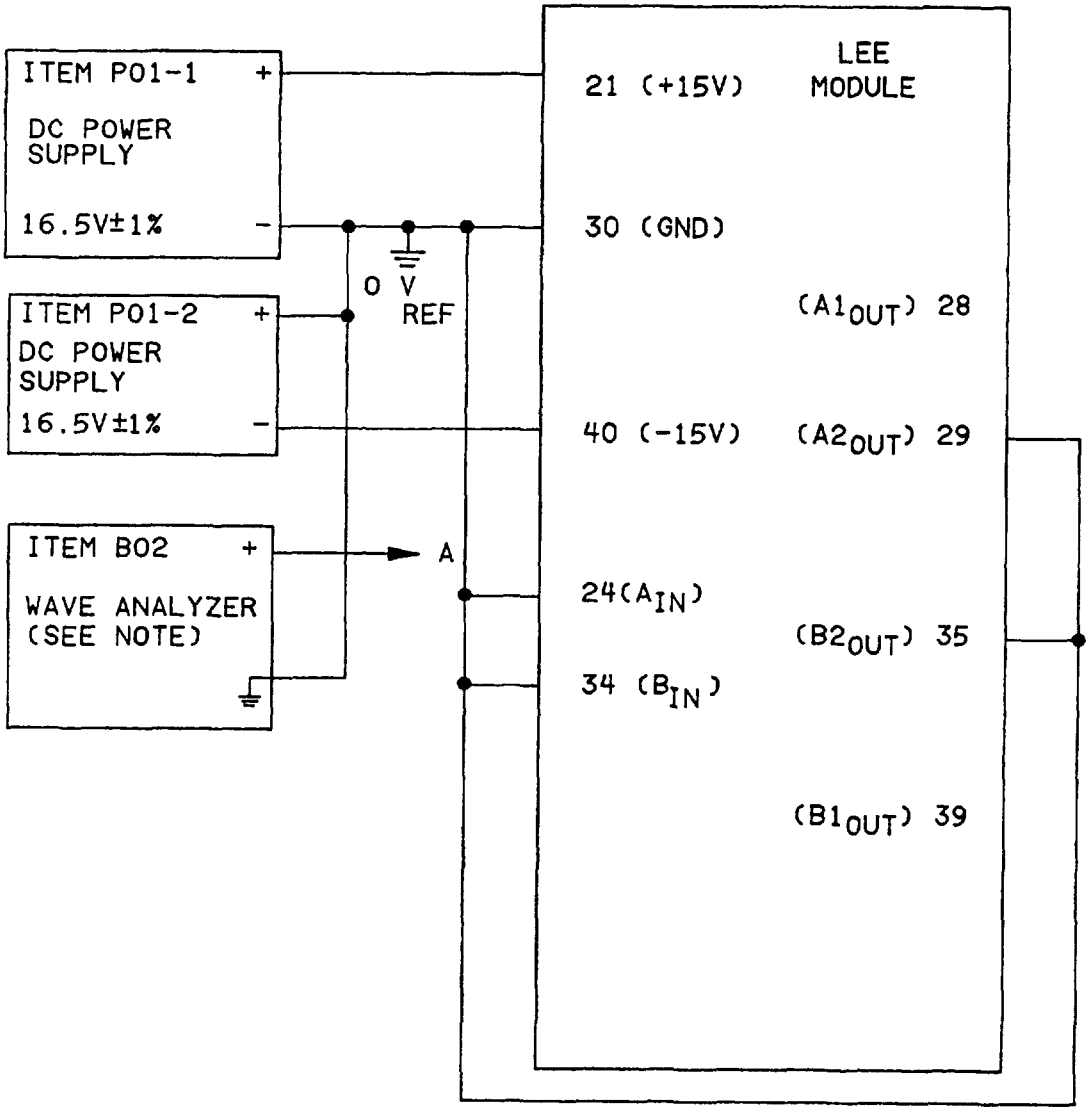
FIGURE 10 Output offset voltage test circuit for table VIII.

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TABLE IX. Output noise test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
H1	Output noise	28	0	20	µV	11
H2	Output noise	39	0	20	µV	11

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NOTE Tune the item B02 across the band, 3 through 20 kHz.
The values measured shall agree with table IX requirements.

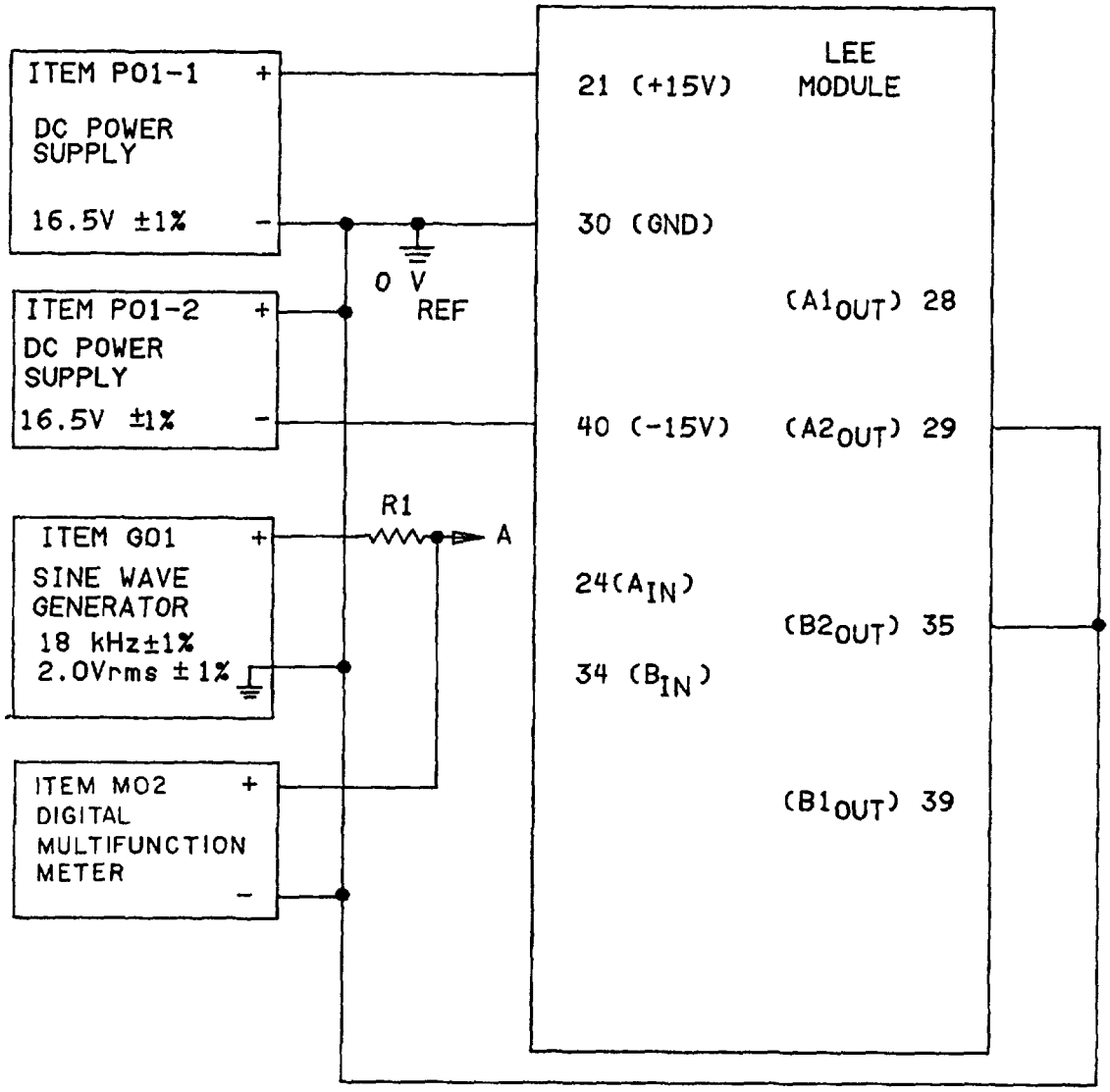
FIGURE 11 Output noise test circuit for table IX

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TABLE X. Input impedance check test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
J1	Input impedance check	24	1.08	1.20	V	12
J2	Input impedance check	34	0.95	1.05	V	12

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NOTE R1 = 2.37 kΩ ±1%, 1/8 W.

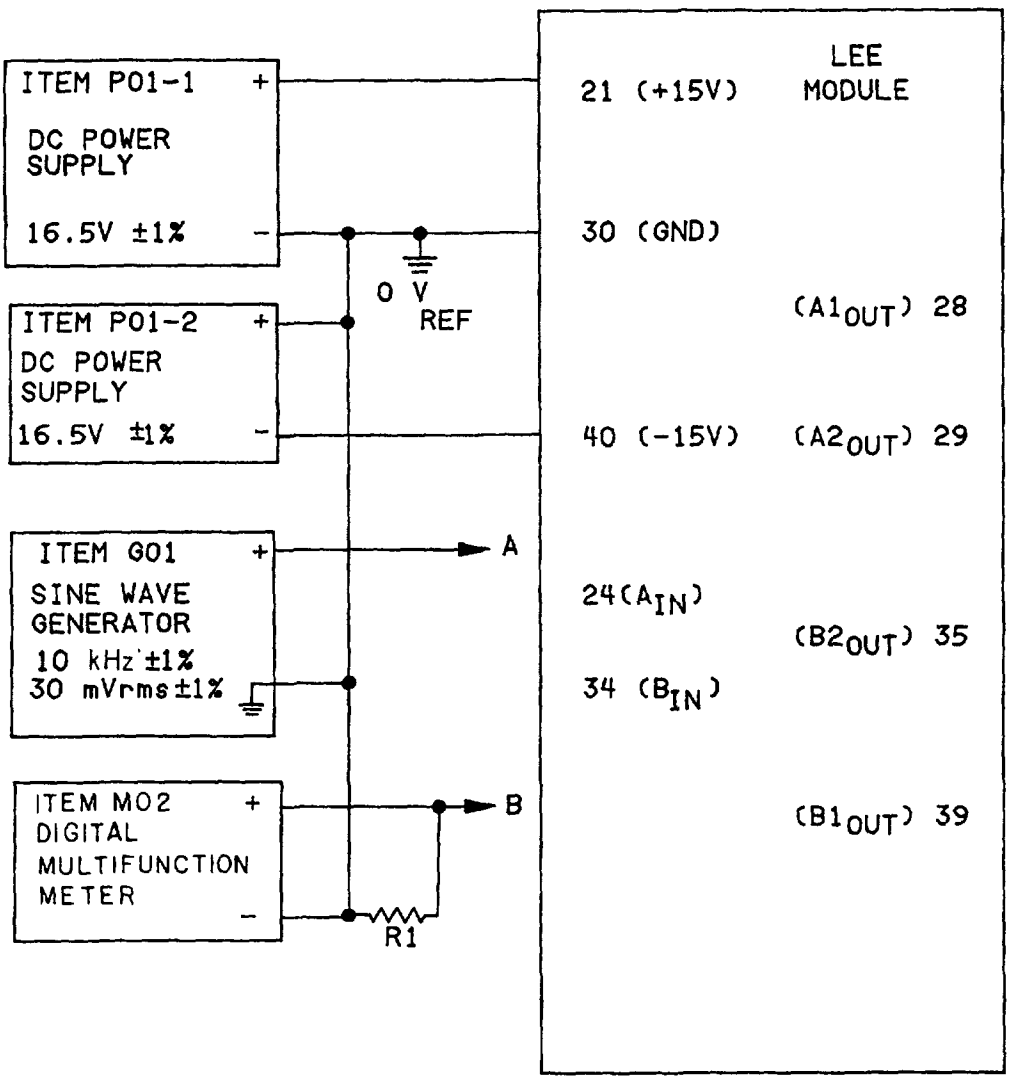
FIGURE 12. Input impedance check test circuit for table X.

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TABLE XI. Output impedance check test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	Output impedance check	24	28	14.25	31.5	mV	13
K2	Output impedance check	34	39	14.25	31.5	mV	13

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NOTE: R1 = 10Ω ±1%, 1/8 W.

FIGURE 13 Output impedance check test circuit for table XI

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TABLE XII. Quiescent power supply current test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
L1	Quiescent power supply current, positive	3.5	16.0	mA	14
L2	Quiescent power supply current, negative	3.5	16.0	mA	14

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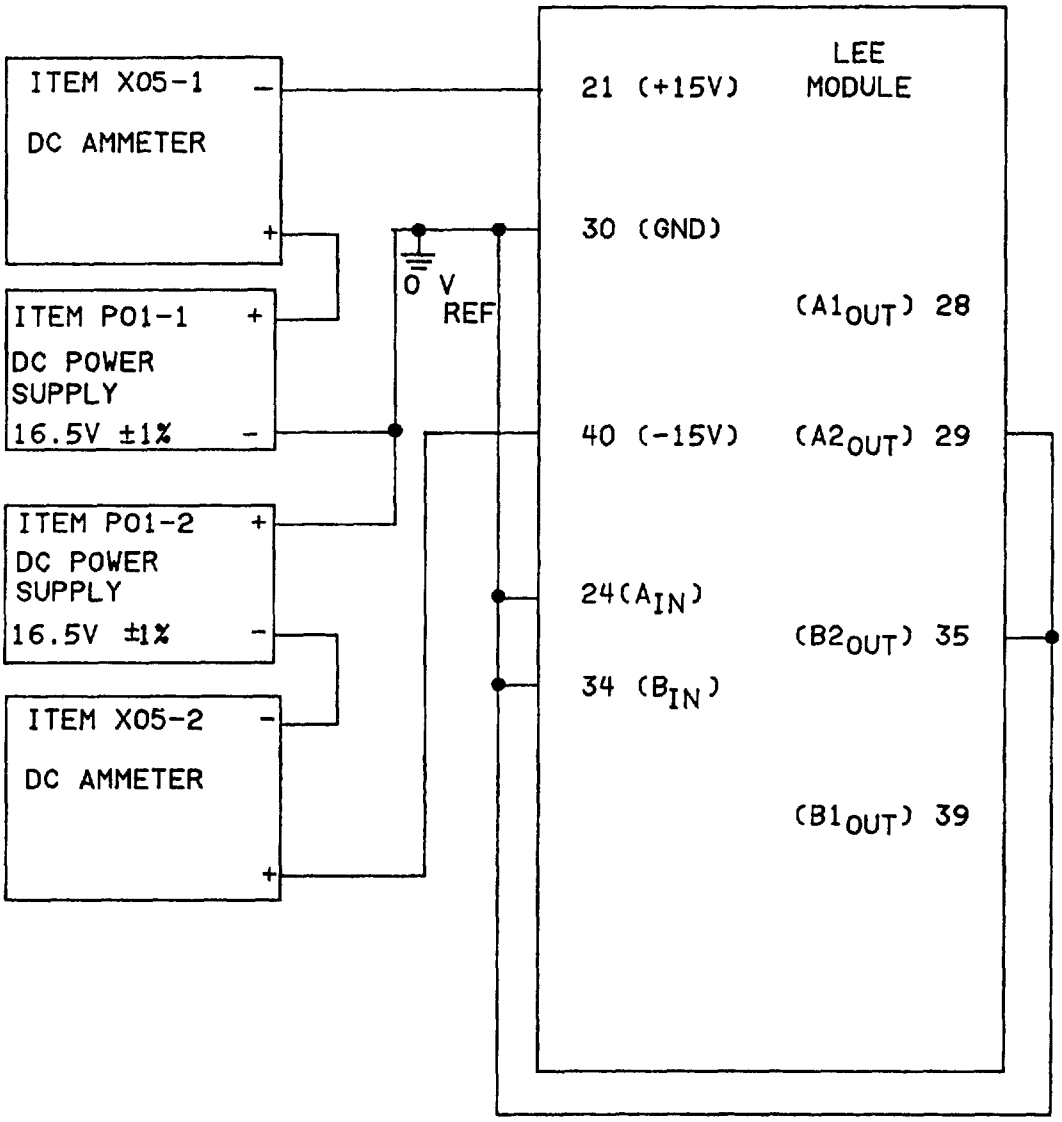


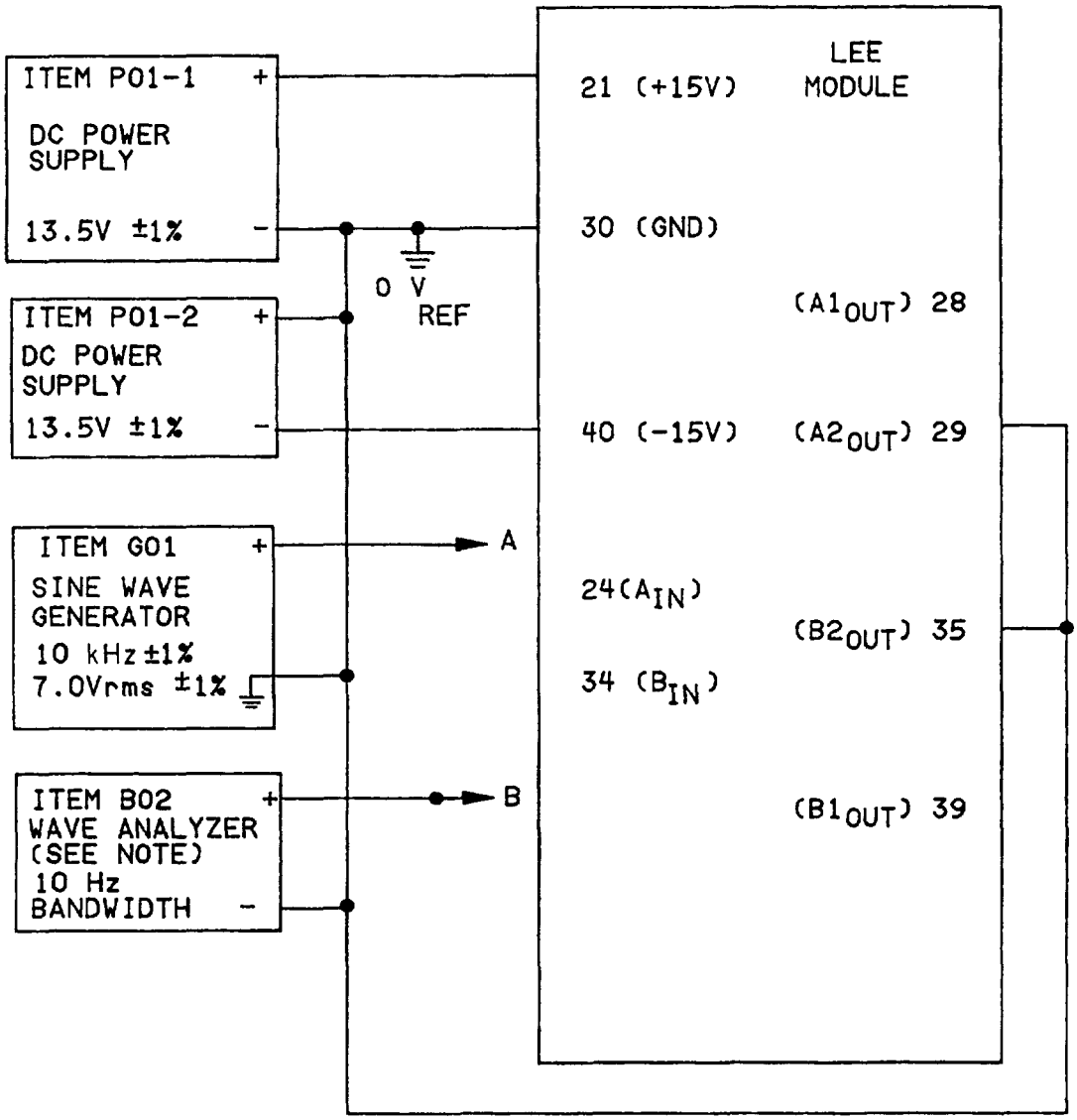
FIGURE 14. Quiescent power supply current test circuit for table XII.

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TABLE XIII. Input common mode voltage swing test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Input common mode voltage swing	24	28	6.65	7.35	V	15
M2	Input common mode voltage swing	34	39	6.65	7.35	V	15

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NOTE Tune item B02 to item G01.

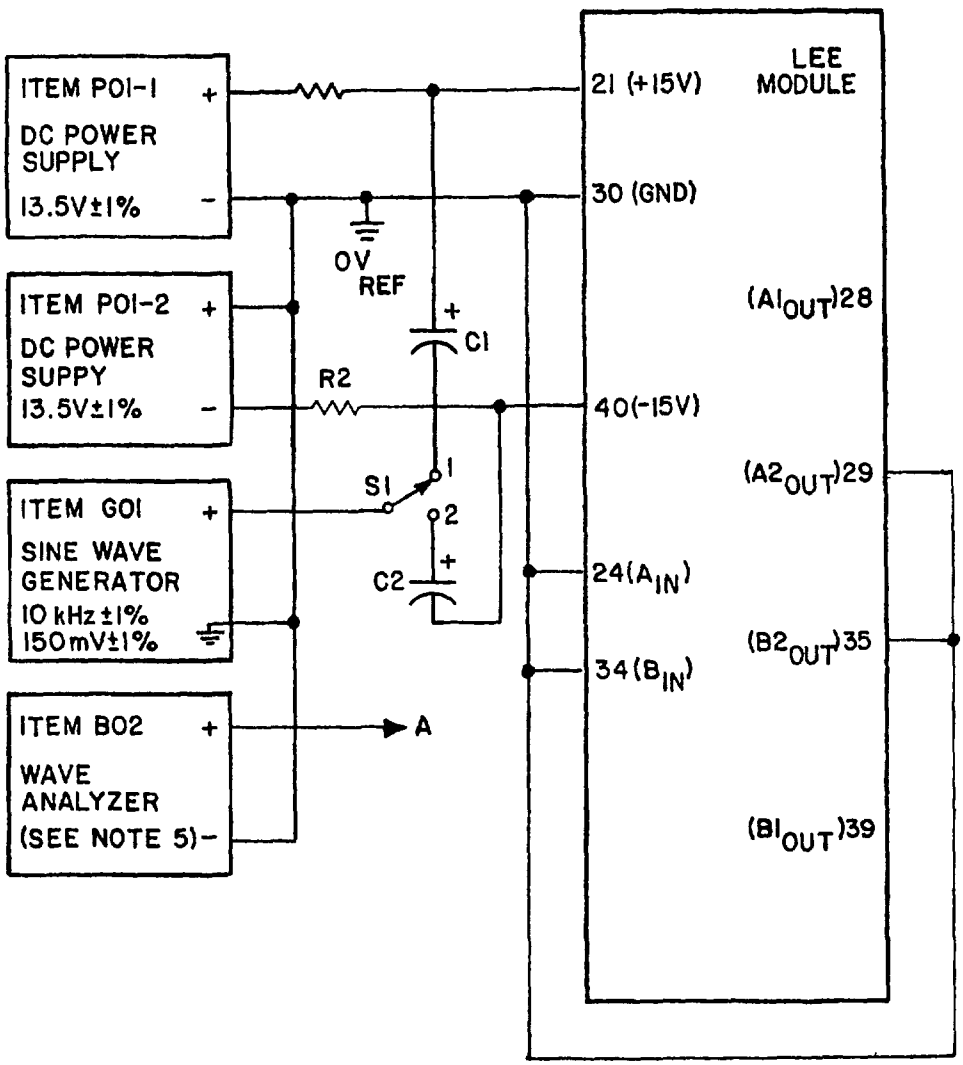
FIGURE 15. Input common mode voltage swing test circuit for table XIII.

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TABLE XIV. Power supply ripple rejection test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
N1	Power supply ripple rejection	28	0	4	mV	16
N2	↓	28	↓	↓	↓	↓
N3		39				
N4		39				

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NOTES:
1. R1 = R2 = 47Ω ±5%, 1/4 W.
2. C1 = C2 = 20 µF ±20%, 20 Vdc.
3. S1 = SPDT switch.
4. Set switch S1 to position 1 for test conditions N1 and N3 and position 2 for test conditions N2 and N4.
5. Tune item B02 to item G01.

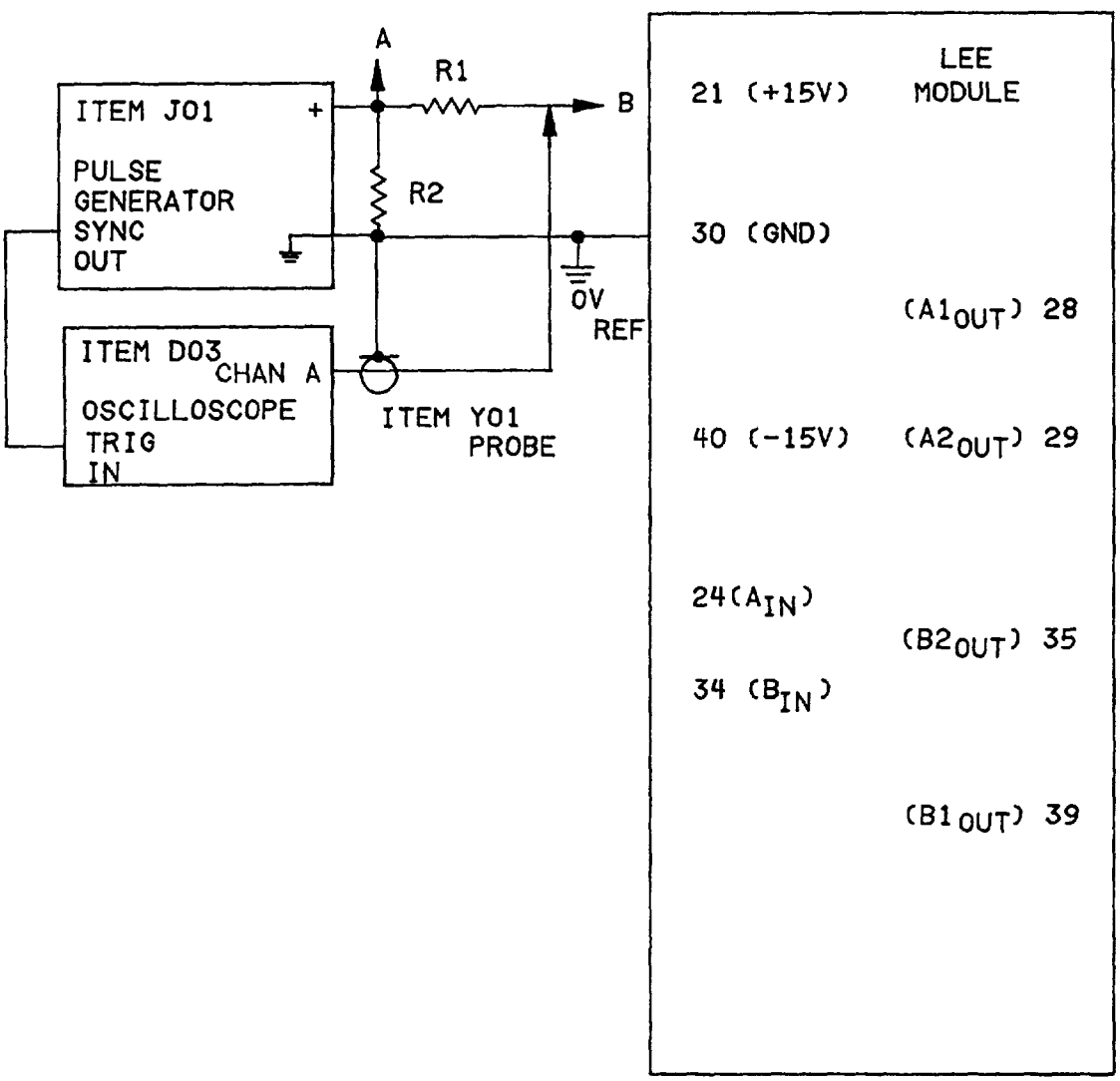
FIGURE 16. Power supply ripple rejection test circuit for table XIV

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TABLE XV. Power supply transient rise time test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
P1	Power supply transient rise time	21	25	75	µs	17,18
P2	Power supply transient rise time	40	25	75	µs	17,18

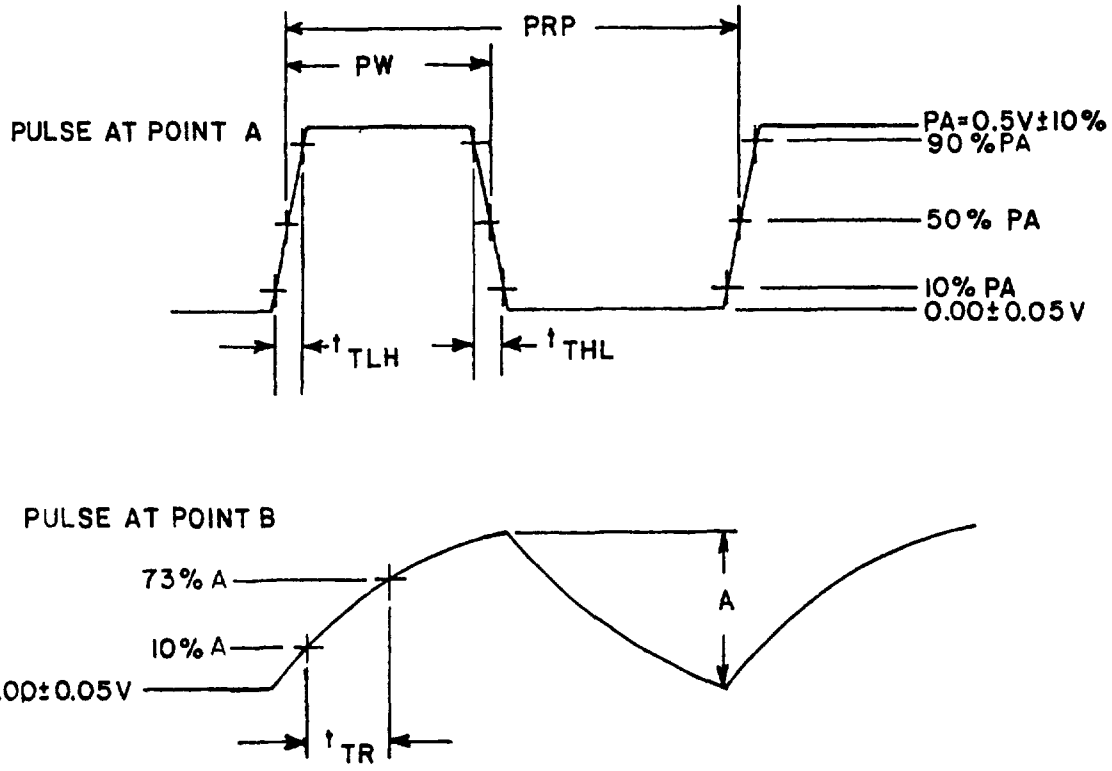
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NOTES:
1. R1 = 412Ω ±1%, 1/8 W
2. R2 = 50Ω ±1%, 1/8 W

FIGURE 17. Power supply transient rise time test circuit for table XV.

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NOTES
1 Input pulse parameters
PW = 300 µs ±10%.
$t_{TLH} = t_{THL}$ = 500 ns maximum
PRP = 600 µs ±10%
2 Measure transient rise time (t_{TR}) at point B.

FIGURE 18. Power supply transient rise time test circuit waveforms for table XV

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